MATH 112B

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1.1 Lecture 1 - Gradient Systems

Notation: for a function with two variables

$$\frac{\partial f}{\partial x} = \lim_{\Delta x \to 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x}$$
$$\frac{\partial f}{\partial y} = \lim_{\Delta y \to 0} \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y}$$

Linearity.

$$\frac{\partial}{\partial x} cf = c \frac{\partial}{\partial x} f \tag{1}$$
$$\frac{\partial}{\partial x}(f + g) = \frac{\partial}{\partial x} f + \frac{\partial}{\partial x} g \tag{2}$$

Gradient. For $f = f(x, y)$, define the gradient as the following:

$$\vec{\nabla} f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right)$$

Properties:

1. Magnitude: $|\vec{\nabla} f| = \sqrt{f_x^2 + f_y^2}$
2. Direction: $\vec{u} = \frac{\vec{\nabla} f}{|\vec{\nabla} f|}$ = unit vector in the $(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y})$ direction.
3. $\vec{\nabla} f$ at point (x_0, y_0) is $\perp$ to the level set at point (x_0, y_0).
4. $\vec{\nabla} f$ is defined at all the points (x, y) where partials exists.
5. A collection of all $\vec{\nabla} f$ is a vector field called gradient field.
6. One could verify $F(M(x, y), N(x, y))$ is a gradient field by verifying

$$\boxed{\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}}$$

   where $F = \vec{\nabla} f$

1.2 Lecture 2 - Conservation Laws

Conservation Laws
This is used to describe: bacterial movement in a petridish, nerve signal traveling through an axon, heat diffusion in a rod, cars on the highway.

We start with the following assumptions:

- Motion happens in 1D.
- Cross-sectional area is constant.

Consider a 1D tube, with particles inside. How do we describe the change of concentration of particles?

$$\begin{array}{c}\text{change in pop between} \\ x \text{ and } x + \Delta x\end{array} = \begin{array}{c}\text{rate of entry into} \\ (x, x + \Delta x) \\ \text{per unit time}\end{array} - \begin{array}{c}\text{rate of exit into} \\ (x, x + \Delta x) \\ \text{per unit time}\end{array}$$

Define

- $c(x, t)$ = concentration of particles at x at time t

- $J(x,t)$ = flux of particles at (x,t) = # of particles crossing a unit area at x in the positive direction, per unit time.
- A = cross-sectional area of the tube.
- ΔV = volume of our segment = $A \cdot \Delta x$

We have

$$\begin{aligned}
\frac{\partial}{\partial t}(\underbrace{c(x,t)\cdot A\cdot\Delta x}_{\text{\# of particles}}) &= J(x,t)A - J(x+\Delta x,t)A \\
\frac{\partial c}{\partial t} &= \frac{J(x,t)-J(x+\Delta x,t)}{\Delta x} \\
\lim_{\Delta x\to 0}\frac{\partial c}{\partial t} &= \lim_{\Delta x\to 0}\frac{J(x,t)-J(x+\Delta x,t)}{\Delta x} \\
\frac{\partial c}{\partial t} &= -\frac{\partial J}{\partial x} \leftarrow \text{Conservation of mass}
\end{aligned}$$

Consider a new term,

$$\begin{matrix}\Delta\text{pop between} \\ x \text{ and } x+\Delta x\end{matrix} = \begin{matrix}\text{rate of entry into} \\ (x,x+\Delta x) \\ \text{per unit time}\end{matrix} - \begin{matrix}\text{rate of exit into} \\ (x,x+\Delta x) \\ \text{per unit time}\end{matrix} \pm \begin{matrix}\text{rate of local} \\ \text{creation} \\ \text{or degradation}\end{matrix}$$

We now have

- $\sigma(x,t)$ = # of particles created/removed per unit volume at time t

Now our conservation laws become

$$\frac{\partial c}{\partial t} = -\frac{\partial J}{\partial x} \pm \sigma(x,t)$$

To generalize to 2D or 3D:

$$\begin{aligned}
\frac{\partial c}{\partial t} &= -\vec{\nabla}\vec{J} \pm \sigma(\vec{x},t) \\
\text{Now we have } c &= c(\vec{x},t) \\
\sigma &= \sigma(\vec{x},t) \\
\vec{J} &= (J_x, J_y, J_z) \\
\vec{\nabla} &= (\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z})
\end{aligned}$$

So we have

$$\frac{\partial c}{\partial t} = -(\frac{\partial J_x}{\partial x} + \frac{\partial J_y}{\partial y} + \frac{\partial J_z}{\partial z}) \pm \sigma(x, y, z, t)$$

$\boxed{\text{ex}}$ Consider a moving fluid + organisms that move with the flow (not actively). Suppose that the velocity of the fluid is given by

$$\vec{v}(x, y, z, t)$$

Let $c(x, y, z, t) =$ concentration of organism.

What is the flux?

$$\vec{J} = c \cdot \vec{V} \quad \text{Assume } \sigma = 0$$
$$\Rightarrow \frac{\partial c}{\partial t} = -\left(\frac{\partial cV_x}{\partial x} + \frac{\partial cV_y}{\partial y} + \frac{\partial cV_z}{\partial z}\right)$$

In 1D:

$$\begin{aligned}\frac{\partial c}{\partial t} &= -\frac{\partial}{\partial x}(cV) \\ &= -\frac{\partial c}{\partial x}V - \frac{\partial V}{\partial x}c\end{aligned}$$

What kind of PDE is this?

- 1st order
- 2 variables: x, t
- Linear
- Transport equation

$\boxed{\text{ex}}$ Attraction/repulsion. Suppose that ψ is a source of attraction for the bacteria. The motion is determined by the gradient of ψ.
In 1D: $\psi = \psi(x, t),\ J = \frac{\partial \psi}{\partial x}$

$$J = c\alpha\vec{\nabla}\psi$$

We have

$$\frac{\partial c}{\partial t} = -\frac{\partial J}{\partial X}$$
$$\frac{\partial c}{\partial t} = -\frac{\partial}{\partial t}\left(c\alpha\frac{\partial \psi}{\psi x}\right)$$

ψ is given.

$$\frac{\partial c}{\partial t} = -\alpha\frac{\partial c}{\partial x}\cdot\frac{\partial \psi}{\partial x} - \alpha c\frac{\partial c}{\partial x}\cdot\frac{\partial^2 \psi}{\partial x^2}$$

This is 1st order, linear PDE in 2 variables.

$\boxed{\text{ex}}$ Temperature of a heated rod.
Here we define $c(x,t)$ as concentration of energy/temperature. And since heat spreads from hot to cold, we get the following.

$$\boxed{\text{Fick's Law} : J = -D\cdot\frac{\partial c}{\partial x}}$$

Then we have

$$\begin{aligned}\frac{\partial c}{\partial t} &= -\frac{\partial}{\partial x}\left(-D\frac{\partial c}{\partial x}\right)\\ &= D\frac{\partial^2 c}{\partial x^2} \;\leftarrow \text{Heat equation}\end{aligned}$$

In higher dimensions:

$$\begin{aligned}\frac{\partial c}{\partial t} &= \vec{\nabla}(D\vec{\nabla}c)\\ &= D\left(\frac{\partial^2 c}{\partial x^2} + \frac{\partial^2 c}{\partial y^2} + \frac{\partial^2 c}{\partial z^2}\right)\end{aligned}$$

$\boxed{\text{ex}}$ Random motion of particles.
Denote λ_r as the amount of particles jumping from $x \to x + \Delta x$
Denote λ_l as the amount of particles jumping from $x \to x - \Delta x$
Denote $c(x,t) = \#$ of particles within $(x, x+\Delta x)$ at time t:

$$c(x, t+\Delta t) = c(x,t) + \lambda_r\cdot c(x-\Delta x, t) + \lambda_l\cdot c(x+\Delta x, t) - \lambda_l c(x,t) - \lambda_r c(x,t)$$

Using Taylor Series:

$$\begin{aligned}
c(x,t+\Delta t) &= c(x,t) + \frac{\partial c}{\partial t}\Delta t + \frac{1}{2}\frac{\partial^2 c}{\partial t^2}(\Delta t)^2 + \dots \\
c(x+\Delta x,t) &= c(x,t) + \frac{\partial c}{\partial x}\Delta x + \frac{1}{2}\frac{\partial^2 c}{\partial x^2}(\Delta x)^2 + \dots \\
c(x-\Delta x,t) &= c(x,t) - \frac{\partial c}{\partial x}\Delta x + \frac{1}{2}\frac{\partial^2 c}{\partial x^2}(\Delta x)^2 + \dots
\end{aligned}$$

We can simplify using $\lambda_r = \lambda_l = \lambda$, then we have

$$\begin{aligned}
c + \frac{\partial c}{\partial t}\Delta t + \frac{\partial^2 c}{\partial t^2}\frac{(\Delta t)^2}{2} + \dots = c& \\
&+ \lambda\left(c + \frac{\partial c}{\partial x}\Delta x + \frac{1}{2}\frac{\partial^2 c}{\partial x^2}(\Delta x)^2\right) \\
&+ \lambda\left(c - \frac{\partial c}{\partial x}\Delta x + \frac{1}{2}\frac{\partial^2 c}{\partial x^2}(\Delta x)^2\right) \\
&- \lambda c - \lambda c \\
\frac{\partial c}{\partial t}\Delta t + \frac{\partial^2 c}{\partial t^2}\frac{(\Delta t)^2}{2} = \lambda\frac{\partial^2 c}{\partial x^2}(\Delta x)^2&
\end{aligned}$$

Call $\lim_{\Delta x\to 0, \Delta t\to 0} \frac{(\Delta x)^2}{\Delta t} \equiv K$ and ignore second order time term as we take the limit.

$$\begin{aligned}
\frac{\partial c}{\partial t}\Delta t &= \lambda(\Delta x)^2 \cdot \frac{\partial^2 c}{\partial x^2} \\
\frac{\partial c}{\partial t} &= \lambda K \cdot \frac{\partial^2 c}{\partial x^2} \leftarrow \text{Diffusion equation}
\end{aligned}$$

1.3 Lecture 3 - Conservation Laws cont.

Recall

$$\frac{\partial c}{\partial t} = -\frac{\partial J}{\partial x} + \sigma(x,t)$$

$\boxed{\text{ex}}$ Propagation of Action Potential along an axon (neural cell). Let

- $x =$ the distance along the axon.

- $q(x,t)$ = the charge density per unit length.
- $J(x,t)$ = flux of charged particles
- $\sigma(x,t)$ = rate of charge entering/leaving

We have the following transport equation for the charge:

$$\frac{\partial q}{\partial t} = -\frac{\partial J}{\partial x} + \sigma(x,t)$$

So what is $J(x,t)$? It is the electrical current.

For $J(x,t)$, we use Ohm's Law:

$$J = -a\frac{\partial V}{\partial x}$$

where a is a constant and $\frac{\partial V}{\partial x}$ is the voltage gradient.

This relates charge to voltage:

$$q - bV$$

where q is charge, b is a constant, V is voltage.

We then have

$$\frac{\partial}{\partial t}(bV) = -\frac{\partial}{\partial x}(-a\frac{\partial V}{\partial x}) + \sigma(x,t)\cdot\frac{1}{b}$$
$$\underbrace{\frac{\partial V}{\partial t} = \frac{a}{b}\frac{\partial^2 V}{\partial x^2}}_{\text{diffusion eq}} + \frac{\sigma(x,t)}{b}$$

Generalize this to 2d or 3d

$$\frac{\partial c}{\partial t} = D\left(\frac{\partial^2 c}{\partial x^2} + \frac{\partial^2 c}{\partial y^2} + \frac{\partial^2 c}{\partial z^2}\right) \qquad \text{(diffusion eq)}$$

Del operator: (note that $\vec{\nabla}u = \text{grad } u$)

$$\vec{\nabla} = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right)$$
$$u(x,y,z) \to \vec{\nabla} \to \left(\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z}\right)$$
$$\text{scalar} \to \vec{\nabla} \to \text{vector}$$

Divergence:
Consider the dot product with $\vec{V} = (V_x, V_y, V_z)$, we have

$$\vec{\nabla} \cdot \vec{V} = \frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z} = \text{divergence of } V$$
$$\text{vector} \to \text{div} \to \text{scalar}$$

Laplacian:
Consider the combination of div and grad

$$\vec{\nabla}(\vec{\nabla} u)$$

where $\vec{\nabla} u = \left(\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z}\right)$. We have

$$\underbrace{\vec{\nabla} \cdot \left(\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial u}{\partial z}\right)}_{\text{div} \cdot \text{grad } u} = \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2}\right) = \underbrace{\Delta u}_{\text{Laplacian } u}$$
$$\Delta = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)$$

Back to the diffusion equation

$$\frac{\partial c}{\partial t} = D\Delta c \qquad \text{(diffusion eq)}$$
$$\frac{\partial c}{\partial t} = D\Delta c + \sigma \qquad \text{(reaction-diffusion eq)}$$

where D is the diffusion coefficient.

This begs the question, what are the units of D?

$$\frac{c}{\text{time}} = D \cdot \frac{c}{\text{space}^2}$$
$$D = \frac{\text{space}^2}{\text{time}}$$

$\boxed{\text{ex}}$ How quickly can a cell get oxygen from blood? (for O_2, $D \approx 0.2\text{cm}^2/\text{sec}$)

1. The approximate distance through which diffusion works in a given time is $\propto \sqrt{Dt}$

2. The approximate time it takes to diffuse a distance d is $\propto \frac{d^2}{D}$

Time it takes O_2 to diffuse

Distance	Diffusion time
1 μm	10^{-3} sec
10 μm	0.1 sec
1 mm	10^3 sec $\approx$ 15 mins
1 cm	10^5 sec $\approx$ 25 hr
1 m	10^8 sec $\approx$ 27 yrs

Role of diffusion in cancer:

Cells in the core of the tumor die due to lack of oxygen, forming a necrotic core. There was a paper by Judah Folkman in 1971, that a tumor can only grow up to 2mm in diameter (physics). This leads to tumors developing methods to deliver oxygen to its core like angiogenesis.

Consider the diffusion equation in 1d:

$$\frac{\partial u}{\partial t} = D\frac{\partial^2 u}{\partial x^2},\ 0 < x < l,\ t > 0$$

We wish to find $u(x,t)$. Set the initial condition to $u(x,0) = f(x)$. Set the boundary condition to $u(0,t) = u(l,t) = 0,\ t > 0$, this is also known as Dirichlet boundary condition.

Alternatively, we could have the Neumann boundary condition, which states that $\frac{\partial}{\partial x}u(0,t) = \frac{\partial}{\partial x}u(l,t) = 0$. This means no flux boundary (insulating).

$\boxed{\text{ex}}$ Consider $\frac{\partial u}{\partial t} = D\frac{\partial^2 u}{\partial x^2},\ 0 < x < 1,\ t > 0$. With initial condition $u(x,0) = f(x) = 2x^2$. With boundary condition $\begin{cases} u(0,t) = 0 \\ u(1,t) = 2 \end{cases}$. Find the long term behavior $\lim_{t\to\infty} u(x,t)$.

<u>Assume</u> that as $t \to \infty$, the solution comes to a fixed profile.

$$\frac{\partial u}{\partial t} = D\frac{\partial^2 u}{\partial x^2} = 0$$

$$\text{As } t \to \infty,\ \frac{\partial u}{\partial t} = 0$$

We have a boundary value problem: $\begin{cases} D\frac{\partial^2 u}{\partial x^2} = 0 \Rightarrow u(x) = cx + b \\ u(0) = 0 \\ u(1) = 2 \end{cases}$

$$\frac{\partial^2 u}{\partial x^2} = 0 \Rightarrow \frac{dV}{dx} = 0 \Rightarrow V(x) = c$$

Use boundary condition:

$$\begin{aligned} u(0) &= c \cdot 0 + b = 0 \Rightarrow b = 0 \\ u(1) &= c \cdot 1 + 0 = 2 \Rightarrow c = 2 \\ \Rightarrow u(x) &= 2x \end{aligned}$$

To solve PDEs, we need additional techniques.

Suppose $f(x)$ defined on $[0, \pi]$. This function can be represented as a sum of sines and cosines.

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx$$

where $b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx dx$ and

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx$$

where $a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx$.
This is known as sine and cosine series representation.

1.4 Lecture 4 -